



# Green(ing) extractivisms: Chile's mining evolution and adaptation to the global energy transition

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## 1. Introduction

This text “locates” Chile in contemporary debates on the global energy transition. It highlights how its focus on electromobility has oversimplified the complexity and multidimensional nature of the current climate crisis. The Chilean case is well known for its historical extractivist logic (Meller, 2007; Salazar and Pinto, 2002). Its extensive sources of renewable energy (primarily solar and wind) and rich mineral ores (such as lithium and copper) are essential for the energy transition. The country is implementing a National Lithium Strategy to participate in (and take advantage of) this transition, giving rise to an updated and novel extractivist logic (one that is distinct from *traditional*, *modern* and *neoextractivism*). This, due to the logic of public-private partnership that is justified as a way of promoting growth and contributing to the global energy transition.

Here we briefly present the evolution of the Chilean case, which serves as a cautionary tale about how an extractive (but updated) logic can masquerade as ‘green’ in similar extractive geographies during the current poly-crisis. The case also emphasizes the critical need for engaging in substantive discussions on fostering a genuinely sustainable societal model. This requires adopting unorthodox, non-productivist, and critically reflective perspectives to ensure the inclusion and representation of diverse voices

The central question guiding this discussion is: Can the destruction of entire ecosystems be justified to enable the exploitation of extractive resources that sustain the consumption patterns of the Global North or finance social policies in resource-rich regions of the Global South, at the expense of both human and non-human life in the Global South? According to the (neo)extractivist logic, and its updated and greener forms, we can; but perhaps we should not.

## 2. Climate reckoning: infinite growth versus a finite world

As predicted since the early 1970s (Meadows et al., 2013), the project of infinitely expanding and hyper-globalized capitalism and the various ecosystems and societies on which it grows and feeds are now in danger of extinction. This situation is a result of the discovery, use, and abuse of fossil fuels (Malm, 2020) extracted from both the coal mines of the global north (Hudson and Beynon, 2021) and the prolific mineral, oil and gas deposits of the south (Imber, 2016; Leff, 2022). Thus, the history of the economic, political and geopolitical geographies that lie between the globalising metropolitan centres and the globalised resource peripheries has been strongly marked by the (neo)liberal project of expanding the indiscriminate exploitation of natural goods and energy resources (Honty and Gudynas, 2014). This expansion guarantees the ongoing growth of production and consumption, preventing Latin America from overcoming the curse that its multiple natural riches have conjured upon it (Galeano, 1971; Humphreys, Sachs, and Stiglitz, 2007).

This context frames the current poly-crisis of civilisation (Bringel and Svampa, 2023), during which socio-economic inequalities in Latin America have deepened. Following the implementation of structural adjustment policies under the Washington Consensus of the 1970s and 80 s and the Commodities Consensus under the ‘Pink Tide’<sup>8</sup> governments at the beginning of the 2000s, today Latin American countries face the Consensus on Decarbonization (Bringel and Svampa, 2023; Svampa, 2013). The latter has been driven by the industrialised economies of the north, where the climate change produced by greenhouse gas emissions can only be addressed by a transition to ‘green energy’ which aims to ensure the standard of living, consumption patterns and electromobility of the elites and other inhabitants of the advanced economies (Dietz, 2023; Dorn, 2019; Svampa and Viale, 2020).

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<sup>8</sup> These were leftist, anti-neoliberal governments that formed a political bloc challenging the Washington Consensus.

There is a consensus that decarbonisation will be achieved through the intensive exploitation of certain minerals by means of energy-intensive processes with stark environmental impacts (Malm, 2020). Extractive production has also been conceptualised as ‘accumulation by defossilisation’; where accumulation is regarded as crucial in the global free-market agendas that foster corporate energy transitions (Slipak & Argento, 2022).

Latin America is of critical importance in this process due to its possession of multiple mineral reserves (such as the lithium and copper used for batteries and solar panels), that are essential to the ongoing energy transition project. However, the deepening of an extractivist logic encouraged by this need for fresh resources has led to the (rein) forcing of multiple processes of deprivation due to its predatory style, extremely high socio-environmental costs, and continuous violation of human and natural rights (Dietz, 2023; Gudynas, 2021; Honty and Gudynas, 2014).

### 3. Evolution of Chilean mining within traditional, modern and new Latin American extractivisms

Like the rest of Latin America, Chile’s history since European colonisation has been one of resource dependence, and so its natural resources have been considered both a blessing and a curse (Auty, 1980; Galeano, 1971). For some Latin American authors, such a history can be understood through the concept of *extractivismo* (extractivism): a long-standing model of appropriation and accumulation which is implemented as a development model based on the exploitation of natural resources by international interests, with little to no value being added to these goods in their countries of origin and significant volumes of them being exported internationally (Gudynas, 2009, 2015b; Svampa, 2019).

This analytical category distinguishes between the *traditional extractivist logic*, associated with the early colonisation processes and the genesis of industrialised Europe, and a *modern extractivism*, whereby extensive international conglomerates control extraction, usually under the protection of neoliberal states. This discussion also identifies new (*neo*)*extractivismos* (neoextractivisms) generated by the Pink Tide governments during the commodity supercycle of the early twenty-first century. During this period, some left-wing Latin American states tightly controlled extractive production by means of nationalisation, expropriation and the creation of state firms to finance urgent and progressive social policies (Gudynas, 2009, 2015b; Riascos, 2014; Svampa, 2019).

Chile was the world’s largest copper producer until the end of the 19th century. It then expanded into Bolivian territory during the Pacific War, inaugurating a traditional extractivist regime characterised by saltpetre mining enclaves, almost absolute control by British capital, and multiple massacres of workers that gave rise to the Latin American Workers’ Movement (Vitale, 2011). After the collapse of the saltpetre industry, the country re-incorporated modern copper extraction led by companies based in the United States (US); this then led to the Chileanisation and nationalisation of the copper industry during the governments of Frei Montalva and Salvador Allende, respectively (Meller, 2007; Salazar and Pinto, 2002). Allende’s *Unidad Popular* (People’s Unity) government would attempt to progressively overcome traditional extractivism under capitalism, advancing towards industrialisation of the country supported by a socialist production system. Yet the nationalisation process would be a critical motivation for the US government’s interference with Chile through the *coup d’état* of 1973, which violently ended such efforts (Salazar and Pinto, 2002).

During the civic-military dictatorship, the country’s dependence on copper deepened. Based on what Naomi Klein calls ‘shock therapy’ (Klein, 2007), the entry of the ‘Chicago Boys’—Chilean economists trained at the University of Chicago by Milton Friedman—into the civic-military dictatorship led to the first implementation of neoliberalism in the world (Harvey, 2005). As Klein (2007) writes, the Chicago

Boys believed that state terrorism was the only way to implement fiscal discipline, tax reforms, trade liberalisation, privatisation and deregulation, and to promote an ideology based on competition and individualism (Springer, 2010). Hence, the civic-military dictatorship inaugurated a tight relationship between extractivism, neoliberalism and the violation of human rights, a relationship which would only deepen during the following decades.

The continuity of this neoliberal extractivist economy would be ensured by the 1980 political constitution established under the dictatorship, which was still in force >30 years after the return to representative democracy in 1990. Since 1990, successive elected governments have notably deepened the neoliberal extractivist model, aggressively opening the Chilean economy to the world through the implementation and expansion of neoliberal policies designed to attract foreign direct investment and keep it flowing into the mining industry (French-Davis, 2018).

During this period, the cracks in the neoliberal extractivist model, especially regarding highly negative externalities in the resource peripheries, intensified. Some of these externalities include the destruction of ecosystems (Weinberg, 2023); water appropriation for mining (in the Atacama Desert, the driest desert in the world) (Mateo-Peinado, 2022); increasingly strained relationships between indigenous communities and the Chilean state (Puschner, 2024; Weinberg, 2023); intense pollution which directly impacts both human health (resource peripheries have the highest cancer rates in Chile and Latin America) and non-human lives (by causing the disappearance of native species) (Tapia et al., 2018; Weinberg, 2023); and stark impacts on urban and rural life (through the long-distance exchange of workers, an increasingly high cost of living, etc.) (Atienza, Arias-Loyola, and Lufin, 2020).

Simultaneously, the 1990s witnessed the widespread implementation of a neoliberal institutional state design, especially in most extractive nations, leading to a lack of regulation under the *laissez-faire* policies of the Washington Consensus (Svampa, 2013, 2019). The extractive nations, avid to lure fresh foreign direct investment, left the free market to solve the massive externalities caused by extractive companies. This led the extractive (mainly mining) companies to promote and implement community development agreements (CDAs) to acquire and maintain a social licence to operate (SLO) from environmentally, socio-economically and culturally affected communities (Boutillier, 2017; Kemp and Owen, 2013; O’Faircheallaigh, 2013).

Hence, the mining industry has fostered models of alleged ‘mutual benefit’ between extractive companies and indigenous and autochthonous communities, taking over the territories and roles abandoned by neoliberal extractive states (i.e., providing infrastructure, education, etc.) (Kemp and Owen, 2013; O’Faircheallaigh, 2013). This has had both superficial and structural long-term impacts on such communities, raising issues like internal friction and fracture, the commodification of land and cultural blurring (Puschner, 2024; Weinberg, 2023); in turn, these issues affect mining companies in a dialectical and dynamic manner.

In all cases such a model has advanced by reason, where extractive firms aim to acquire and retain an SLO by fostering CDAs which lead to “bread for today, hunger for tomorrow” (as cited in Puschner, 2024), imposing this by force<sup>h</sup> through the violation of human rights and the rights of nature (Poo, 2023). Under Chilean neoliberalism, frictions between the extractive industries and civil society are usually ironed out through state repression in a context of increasing awareness of issues relating to the environment, the climate and human dignity (Leff, 2014; Poo, 2023).

### 4. The Chilean model shows its cracks

All the above, coupled with an increasing socio-economic inequality,

<sup>h</sup> “By reason or by force” is the Chilean coat of arms.

led to the Social Outbreak of 2019. Over the course of several weeks, millions demanded a decent and sustainable life for human and non-human beings; the genuine inclusion of gender diversity/dissent, along with indigenous and migrant peoples; and the construction of a sociopolitical system based on an ethos of solidarity and mutual aid—all of which are unimaginable under neoliberal extractivism (Arias-Loyola, 2021). The protests were violently suppressed by the state, leading to human rights violations across the country, as the United Nations Human Rights Office (2019) recognised. State institutions quickly reacted to dismantle the protests, leading to political parties signing an ‘Agreement for Peace’ which set in motion a referendum to determine whether a new constituent process would begin. In parallel, the COVID-19 pandemic hit the country, forcing an end to the protests.

The referendum took place in 2020, and the writing of a new draft constitution was approved. This was the first constitutional process in the world to elect constituents according to gender parity, and also incorporated special quotas for the different indigenous ethnicities recognised to date (Arias-Loyola, 2021). The constitutional process generated spaces where broad issues could be discussed, such as the rights of nature (Gudynas, 2015a), strategies for overcoming extractivism, the possibility of non-capitalistic production and degrowth (Leff, 2019), and how the poly-crisis can be addressed through a systemic transition (energetic, economic, sociopolitical and cultural) towards a society that is capable of ensuring a dignified life in balance with (and not at the cost of) the ecosystem.

The draft proposed by the Constitutional Convention was considered the “world’s most progressive constitution” by parts of the international press (Bartlett, 2002). Still, it was ultimately rejected in a process heavily laden with misinformation, social polarisation, and mutual accusations. This pivotal moment in the nation’s history brings to mind Marx’s famous phrase, that history repeats itself first as tragedy and then as farce: a second attempt to redraft the constitution, this time led by the forces of the right wing and based on a perspective firmly rooted in the hegemonic point of view, was once again rejected.

While this was happening, the former student leader Gabriel Boric was elected president of Chile. In line with the first constitutional draft, Boric’s programme envisaged the installation of a ‘feminist and environmentalist’ government which would move towards overcoming the logic of neoliberal extractivism. However, the crushing defeat of the first constitutional draft led Boric’s government to drastically change its (dis)course. Lacking state funds and public support, Boric’s government decided to promote several extractive-related policies; these included the first *mining royalty*, the possibility of exporting green hydrogen and—most importantly—the National Lithium Strategy (ENL). This is based on the call made by the Intergovernmental Panel of Experts on Climate Change (IPCC) to move towards energy models which are resilient to climate change.

## 5. Greening Chilean neoliberalism extractivism?

The ENL states that it aims to “surpass extractivist logic”, preventing the lithium industry from generating new forms of enclave. It also proposes to continue and intensify the exploitation of the *Salar de Atacama* (the Atacama Salt Flat, the world’s largest lithium operating site), to keep exploring other salt flats for the purpose of lithium exploitation, and to promote the generation of value through forward and backwards linkages in a sustainable and community-sensitive manner. The goal is to foster national development through the export of lithium while implementing a novel public-private partnership model whereby the Chilean state will have the majority stake in the new National Lithium Company.

In other words, the ENL frames itself within the discussion of post-extractivism (Gudynas, 2011), but ends up promoting a novel form of extractivism that is distinct from previously identified types. This cannot be called neoextractivism, since the state does not aim to replace the private extraction of minerals as the Pink Tide governments did. Nor can

it be considered a modern form orthodox extractivism, as it does not prioritise the promotion of free markets and the complete control of extraction by international capital. The ENL is thus an innovation that blends features of neoextractivism and extractivism, as the state aims to maintain majority control of the extractive firm in order to finance social and economic policies. At the same time, it partners with private companies through competitive tenders under a neoliberal framework.

This novel form of extractivism reverses the concepts established during the Chilean Social Outbreak and its first Constituent Process, and also several principles advanced in the broader Latin American context, like the rights of nature (Gudynas, 2015a), the invisibility of gender in energy transitions (Rátiva-Gaona and Rátiva-Gaona, 2023), and the value of environmental knowledge (Leff, 1998). It focuses on the production and enhancement of value rather than sustainable value capture within (or in favour of) resource peripheries. This is a vital point, since the value produced and enhanced in such peripheries is mostly captured by global metropolises both within and outside the country. Furthermore, this updated model is built under the hypothesis that forward and backward socio-productive linkages will be spontaneously generated, leading to the productive diversification of the resource peripheries. Worryingly, it also borrows elements from neoextractivism, such as the dependence of social policies on extractive industries and the consequent danger of fostering an economic populism that is based on projects of infinite economic growth.

This new mode of extractivism is ‘painted green’, but seems as unsustainable as its older siblings. It is currently being validated through a discourse of paternalistic goodwill in which countries like Chile gather and sign multiple treaties and conventions which involve little (if any) serious commitments. Some examples relate to improving the incorporation of indigenous voices (ILO Convention 169), safeguarding biological diversity, human rights and the environment (Convention on Biological Diversity, Ramsar Convention, Escazú Agreement) and dealing with climate change (United Nations Framework Convention on Climate Change, Kyoto Protocol and Paris Agreement). All these agreements aim to contribute towards the decarbonisation of capitalist relations while completely ignoring greenhouse gases other than carbon dioxide and projects of transition towards non-capitalistic societies (Poo, 2023; Svampa, 2020).

Paradoxically, the extraction of minerals needed for the energy transition towards green(er) sources of energy, especially in terms of electromobility in the global north, implies the increasing exploitation of the same environments that this transition – allegedly – aims to safeguard. Hence, this novel and greener form of extractivism could also be considered an updated model in which the resources required by the global north for the energy transition are accumulated through dispossession of the environment in the global south. If this model successfully disguises a rentist logic as green development, this raises concerns about whether continued extractive production will keep reproducing the unsustainable conditions that precipitated the climate crisis in the first place.

In linking the discourse on green extractivism in Chile with the natural sciences, particularly ecology and conservation biology, we unearth pivotal intersections. The notion of ‘green’ extractivism contains an essential contradiction: while purportedly serving the global north’s transition to renewable energy, it perpetuates ecological degradation. This paradox becomes more palpable when viewed through the lens of ecosystem resilience and species conservation. Extensive mining operations in biodiverse regions, such as the Atacama Desert, highlight the urgent need for holistic approaches that integrate geochemical and biological data to comprehensively assess environmental impacts. Engaging with ecological theories on habitat fragmentation and edge effects could provide deeper insights into the systemic transformations required to mitigate the adverse consequences of extractive activities. Thus, an interdisciplinary framework linking the methodologies used in social and natural sciences could pave the way for genuinely sustainable mining practices that align more closely with

conservation goals, ensuring that the pursuit of green energy does not come at the cost of ecological integrity.

### CRediT authorship contribution statement

**Martín Arias-Loyola:** Writing – review & editing, Writing – original draft, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Francisco Vergara-Peruich:** Writing – review & editing, Writing – original draft, Resources, Investigation. **Felipe Encinas:** Writing – original draft, Investigation.

### Conflicts of interest

None.

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